

Advanced (Habanero)

Q1. Solve the equation $x^2 + 5x + 6 = 0$ by factoring.

- (A) $x = -2, x = -3$
- (B) $x = -1, x = -6$
- (C) $x = 2, x = 3$
- (D) $x = 1, x = 6$
- (E) $x = -2, x = 3$

Q2. Solve the equation $x^2 - 7x + 12 = 0$ by factoring.

- (A) $x = 3, x = 4$
- (B) $x = 3, x = 12$
- (C) $x = 7, x = 12$
- (D) $x = 4, x = 12$
- (E) $x = 3, x = 5$

Q3. Solve the equation $x^2 + 2x - 6 = 0$ by factoring.

- (A) $x = -1 \pm \sqrt{7}$
- (B) $x = 1 \pm \sqrt{7}$
- (C) $x = -3, x = 2$
- (D) $x = -2, x = 3$
- (E) $x = -1, x = 6$

Q4. What is the solution to the equation $x^2 - 9 = 0$?

- (A) $x = \pm 3$
- (B) $x = \pm 9$
- (C) $x = 3$
- (D) $x = -3$
- (E) $x = 0$

Q5. Solve the equation $x^2 + 10x + 24 = 0$ by factoring.

- (A) $x = -4, x = -6$
- (B) $x = -4, x = 6$
- (C) $x = 4, x = 6$
- (D) $x = -2, x = -12$
- (E) $x = 2, x = 12$

Q6. Solve the equation $x^2 - 4x - 5 = 0$ by factoring.

- (A) $x = -1, x = 5$
- (B) $x = 1, x = -5$
- (C) $x = 5, x = -1$
- (D) $x = -1, x = -5$
- (E) $x = 1, x = 5$

Q7. Solve the equation $x^2 + x - 12 = 0$ by factoring.

- (A) $x = -4, x = 3$
- (B) $x = -3, x = 4$
- (C) $x = 3, x = -4$
- (D) $x = -3, x = -4$
- (E) $x = 4, x = 3$

Q8. Solve the equation $x^2 - 2x - 15 = 0$ by factoring.

- (A) $x = -3, x = 5$
- (B) $x = 3, x = -5$
- (C) $x = 5, x = -3$
- (D) $x = 3, x = 5$
- (E) $x = -5, x = -3$

Open Ended Questions (FRQ)**Q9.** Solve the equation $x^2 + 4x + 4 = 0$ by factoring and verify the solution.**Q10.** Solve the equation $x^2 - 5x - 6 = 0$ by factoring and verify the solution.**Chef's Tips**

- Tip 1: Make sure students show their work and explain their reasoning.
- Tip 2: Remind students to check their solutions by plugging them back into the original equation.
- Tip 3: Encourage students to use factoring to solve the equations whenever possible.